

RA-1 Semantic Unit Class & Runtime Admissibility Architecture

Technical Documentation – v1.0

1. Overview

The RA-1 semantic unit class is the foundational construct used by SSA to represent institutional meaning, authority applicability, conflict states, and runtime admissibility. Unlike rule-based governance systems, RA-1 treats authority as a *dynamic semantic state* rather than a static entitlement. This allows the system to determine whether an action is institutionally admissible **at the moment of execution**, based on current conditions.

2. Semantic Unit Structure (RA-1)

Each RA-1 unit contains:

- **Validity Period** Temporal boundaries within which the unit may be applied. Represented as: `valid_from`, `valid_until`, `temporal_dependencies`.
- **Jurisdiction Scope** The institutional or geographic domain in which the unit holds authority. Represented as: `jurisdiction_id`, `subordination_relations`, `cross-domain constraints`.
- **Dependent Evidence** External or internal facts required for the unit to remain admissible. Represented as: `evidence_ref`, `evidence_state`, `evidence_expiry`.
- **Delegation Boundaries** The limits of authority transfer. Represented as: `delegation_source`, `delegation_scope`, `delegation_constraints`.
- **Semantic Relations** Logical and institutional relationships between units. Represented as: `supports`, `contradicts`, `overrides`, `requires`, `suspends`.

These elements allow RA-1 units to evolve asynchronously without invalidating the entire governance structure.

3. Conflict Representation (Semantic Antinomy)

Conflicts are not treated as rule failures. They are represented as **semantic antinomies**, each with:

- **Type** (e.g., jurisdictional conflict, temporal conflict, evidentiary conflict)
- **Origin** The semantic unit or dependency that produced the contradiction.
- **Resolution Direction** The deterministic path the system should take: `escalate`, `suspend`, `override`, `defer`, `block`.

This allows the system to preserve the *meaning* of the conflict and act predictably.

4. Registry Structure

The SSA registry maintains all RA-1 units and their current states. It includes:

- **Unit State Table** Tracks validity, jurisdiction, evidence, and delegation in real time.
- **Dependency Graph** Represents semantic relations and conflict paths.
- **Admissibility Index** A computed value indicating whether an action may proceed.
- **Evidence Ledger** Immutable record of admissibility decisions and their semantic causes.

The registry is designed for deterministic evaluation and auditability.

5. Runtime Admissibility Evaluator

At execution time, the evaluator checks:

1. **Temporal validity** Has the unit moved outside its valid period?
2. **Jurisdictional alignment** Has the jurisdiction changed or become incompatible?
3. **Evidentiary state** Has required evidence expired, changed, or become contradictory?
4. **Delegation boundaries** Is the delegated authority still within scope?
5. **Semantic conflicts** Have any units entered antinomy?

If any condition transitions outside its admissible range, the action becomes **non-admissible**, even if identity and delegation are correct.

6. Evidence Preservation

After each admissibility decision, the system records:

- the semantic units involved,
- their states at the moment of evaluation,
- any conflicts detected,
- the resolution direction,
- the final admissibility outcome.

This creates a deterministic, reconstructable path explaining *why* an action was allowed or blocked at that moment.

7. Research Implications

The SSA architecture suggests that a sufficiently expressive semantic layer can provide runtime admissibility **without requiring a separate governance mechanism**. The boundary between semantic governance and admissibility evaluation becomes a matter of representation, not system separation.

This directly aligns with ongoing research into dynamic institutional conditions and action-time governance.

8. Next Steps

Full RA-1 specification and registry schema can be provided upon request. Further documentation includes:

- RA-1 Formal Specification
- Registry Schema v1.2
- Semantic Conflict Resolution Guide
- Runtime Admissibility Evaluation Model
- Evidence Ledger Format



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